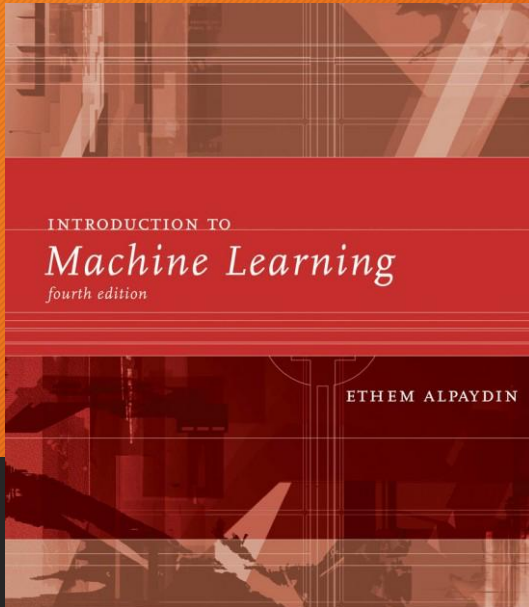




Lecture Slides for

INTRODUCTION TO Machine Learning

4e

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CHAPTER 2:

Supervised Learning

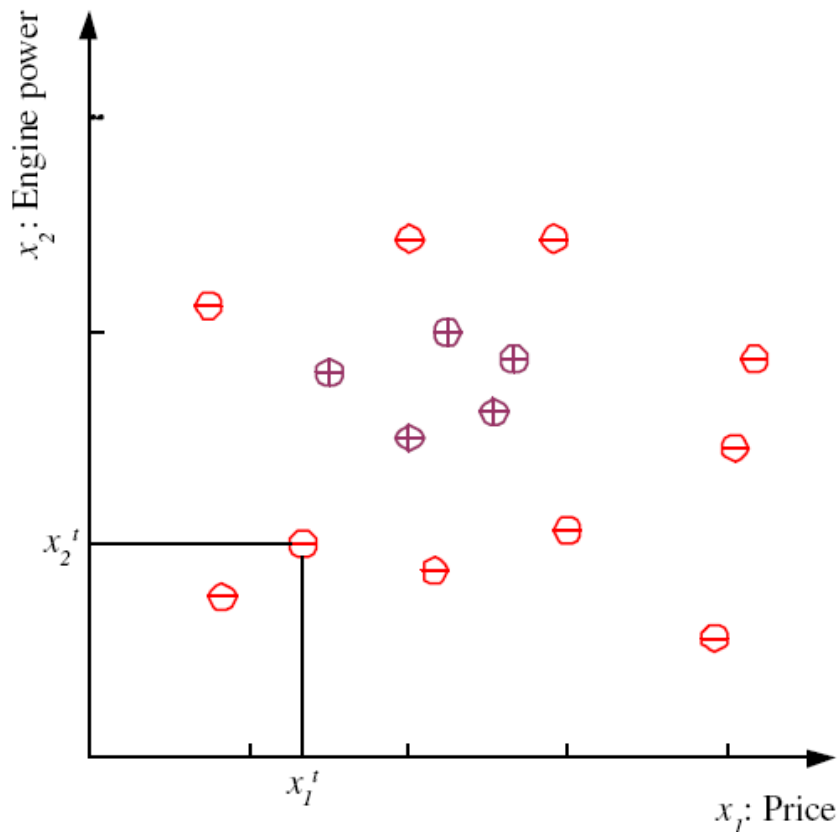
Learning a Class from Examples

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- Class C of a “family car”
 - **Prediction:** Is car x a family car?
 - **Knowledge extraction:** What do people expect from a family car?
- Output:
 - Positive (+) and negative (-) examples
- Input representation:
 - x_1 : price, x_2 : engine power

Training set $\mathcal{X}$

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$$\mathcal{X} = \{\mathbf{x}^t, r^t\}_{t=1}^N$$

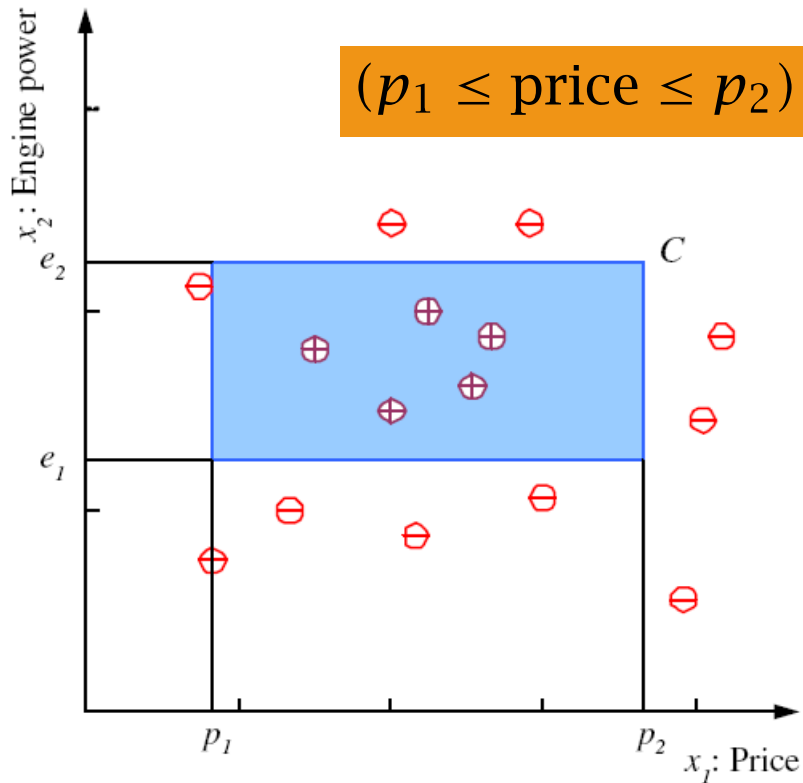
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$r = \begin{cases} 1 & \text{if } \mathbf{x} \text{ is a positive example} \\ 0 & \text{if } \mathbf{x} \text{ is a negative example} \end{cases}$$

Class C

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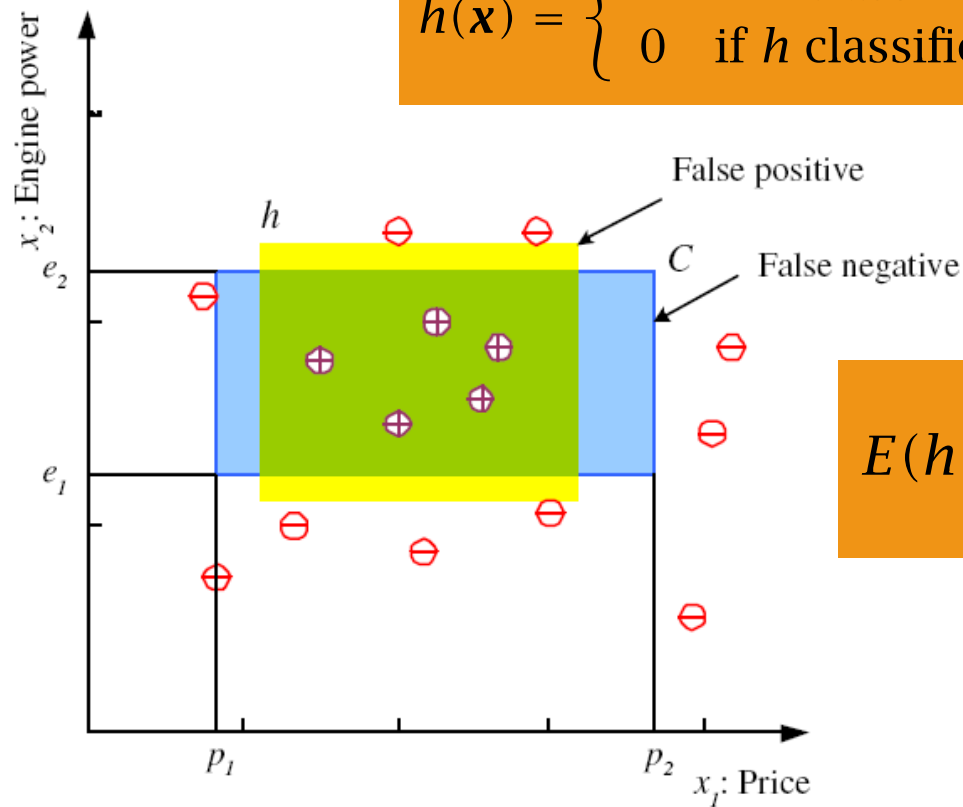
$(p_1 \leq \text{price} \leq p_2) \text{ AND } (e_1 \leq \text{engine power} \leq e_2)$



Hypothesis class $\mathcal{H}$

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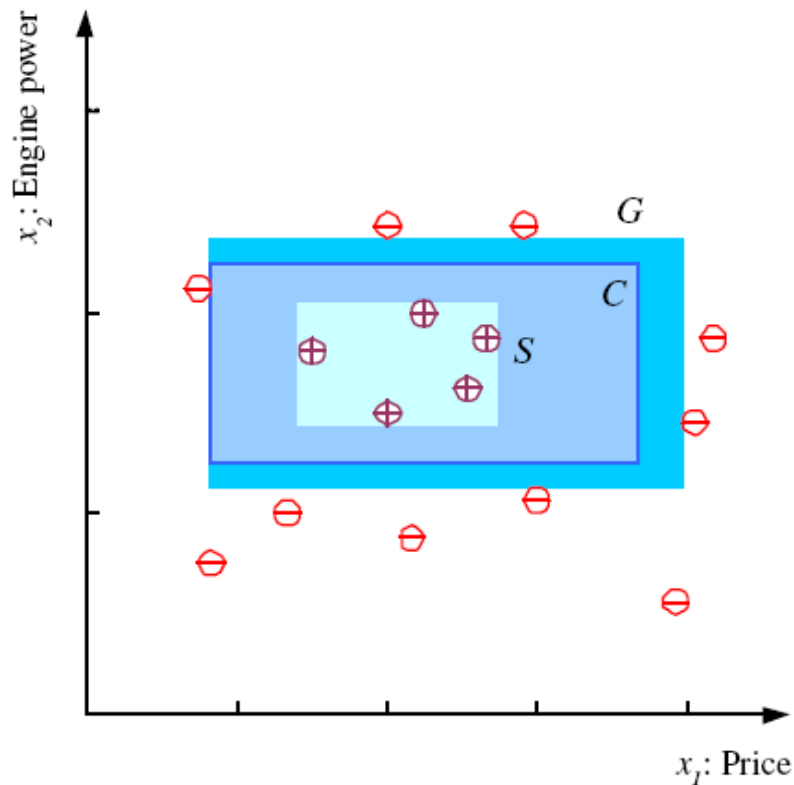
$$h(\mathbf{x}) = \begin{cases} 1 & \text{if } h \text{ classifies } \mathbf{x} \text{ as a positive example} \\ 0 & \text{if } h \text{ classifies } \mathbf{x} \text{ as a negative example} \end{cases}$$



$$E(h|\mathcal{X}) = \sum_{t=1}^N 1(h(\mathbf{x}^t) \neq r^t)$$

S, G, and the Version Space

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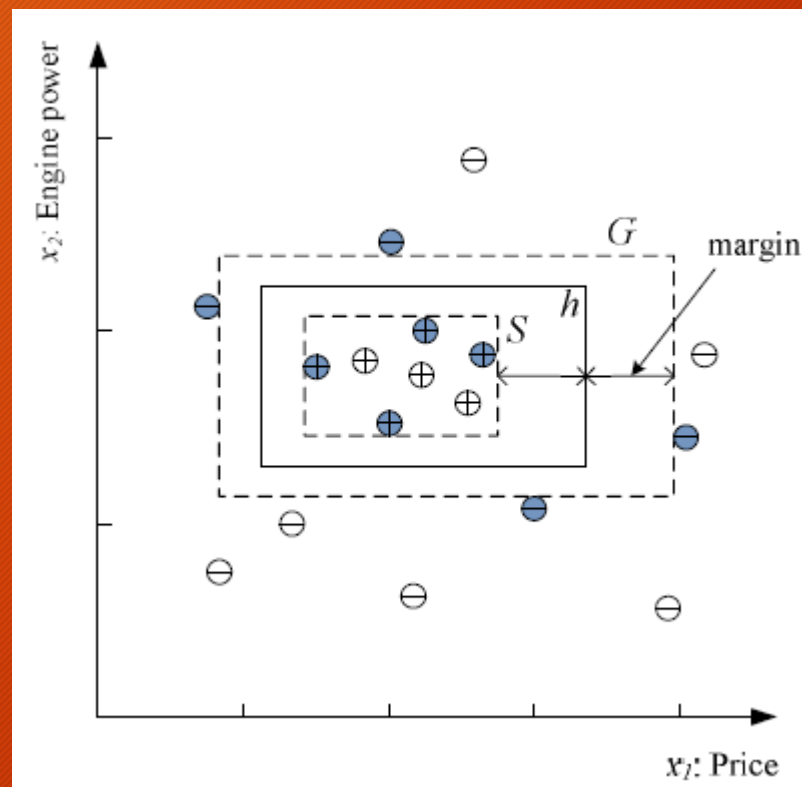


S , the most specific hypothesis,
 G , the most general hypothesis,
and any $h \in H$ between S and G
is **consistent** and make up the
version space
(Mitchell, 1997)

Margin

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- Choose h with largest margin

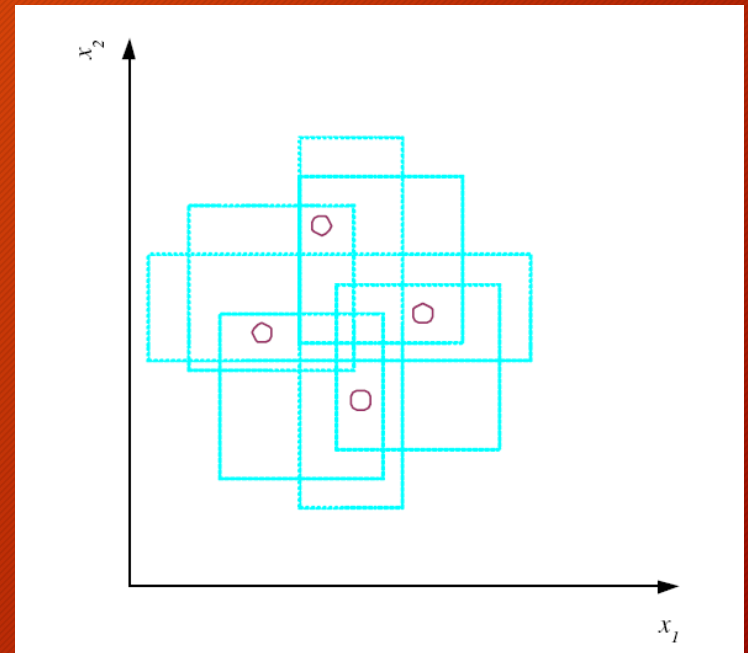


VC Dimension

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- N points can be labeled in 2^N ways as $+/-$
- $\mathcal{H}$ **shatters** N if there exists $h \in \mathcal{H}$ consistent for any of these:
$$VC(\mathcal{H}) = N$$

An axis-aligned rectangle shatters 4 points only !



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- (Blumer et al., 1989)

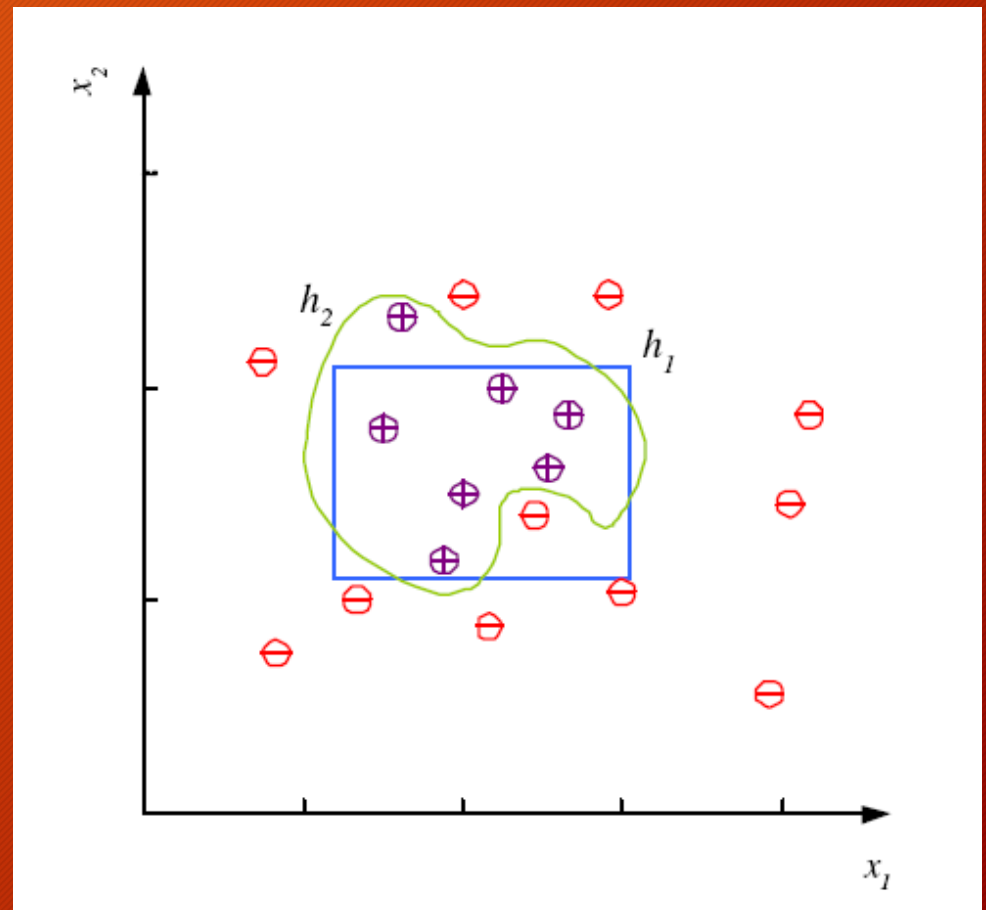
-
- The diagram shows a 2D coordinate system with axes x_1 and x_2 . A central cyan rectangle is labeled h . It is surrounded by a blue region, and a purple horizontal bar is on top of the blue region. Red circles with '+' and '-' signs are scattered around. The label C is next to the purple bar.

Noise and Model Complexity

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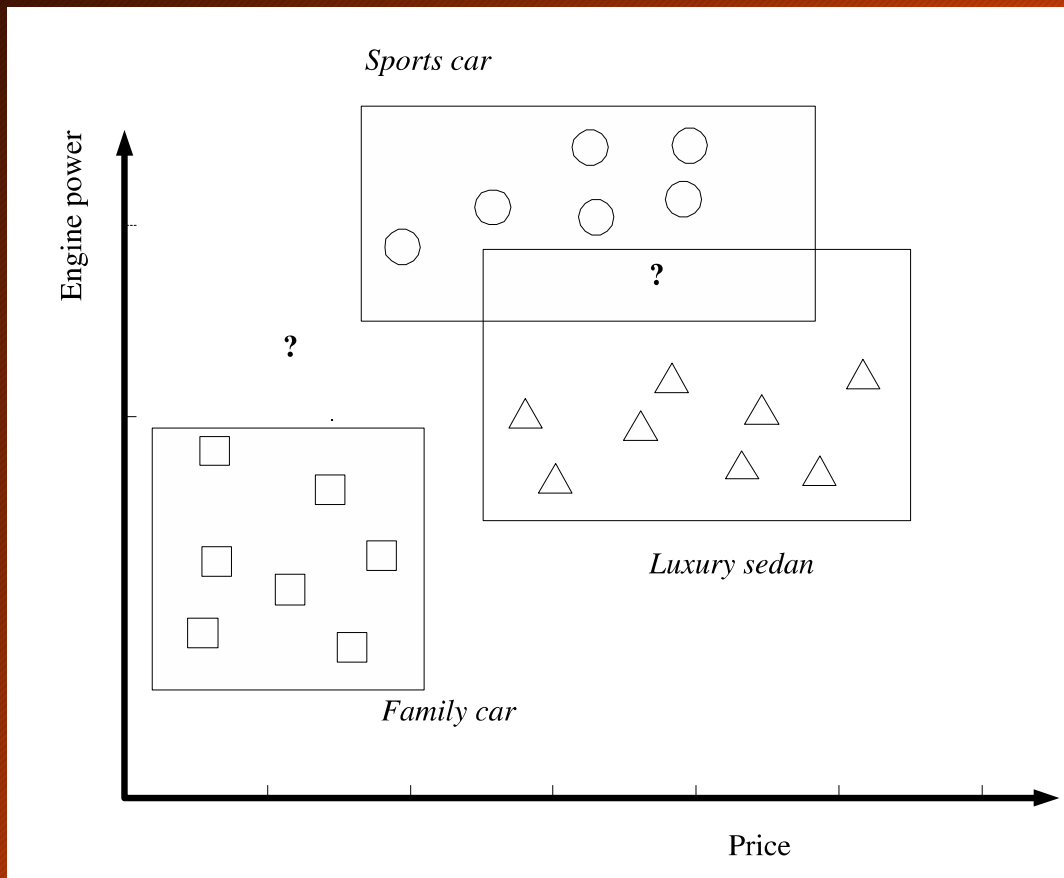
Use the simpler one because

- Simpler to use (lower computational complexity)
- Easier to train (lower space complexity)
- Easier to explain (more interpretable)
- Generalizes better (lower variance - Occam's razor)



Multiple Classes, C_i $i=1,\dots,K$

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Train hypotheses $h_i(\mathbf{x})$, $i=1,\dots,K$, one to separate each class from the union of all others

?: reject regions where no or more than one $h_i(\mathbf{x}) > 0$.

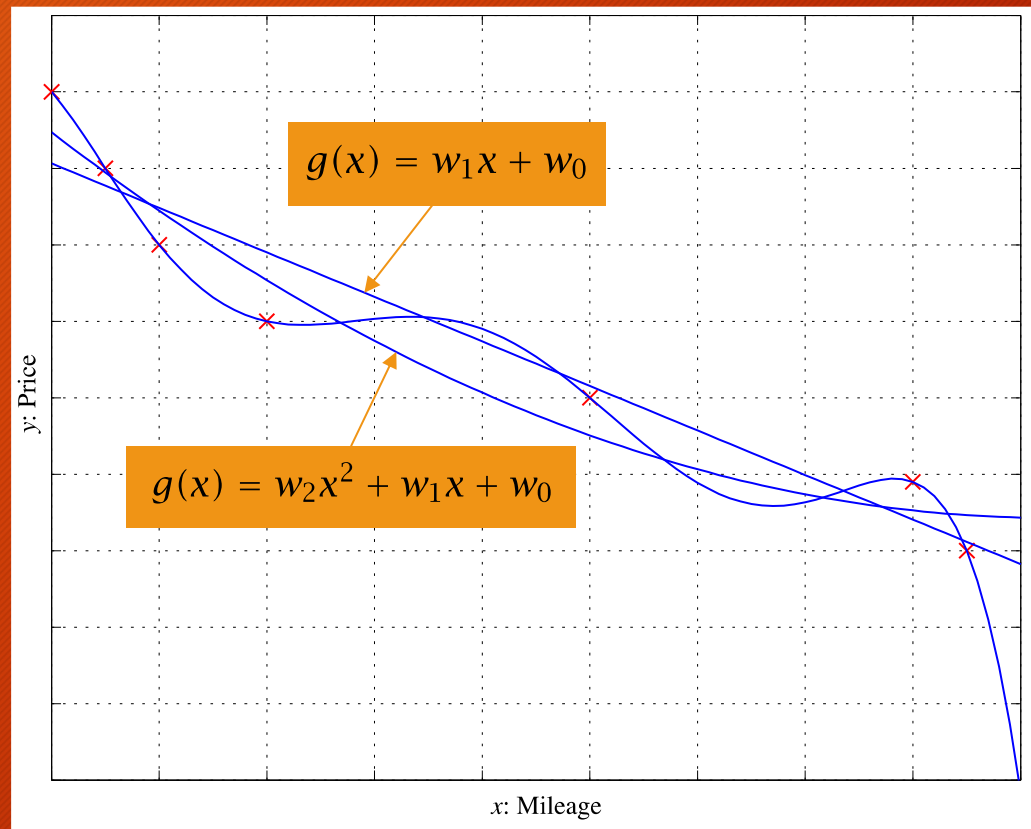
Regression

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$$\mathcal{X} = \{\mathbf{x}^t, r^t\}_{t=1}^N$$

where $r^t \in \mathbb{R}$.

$$E(g|\mathcal{X}) = \frac{1}{N} \sum_{t=1}^N [r^t - g(\mathbf{x}^t)]^2$$



Model Selection & Generalization

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- Learning is an **ill-posed problem**; data is not sufficient to find a unique solution
- The need for **inductive bias**, assumptions about $\mathcal{H}$
- **Generalization**: How well a model performs on new data
- Overfitting: $\mathcal{H}$ more complex than C or f
- Underfitting: $\mathcal{H}$ less complex than C or f

Triple Trade-Off

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- There is a trade-off between three factors (Dietterich, 2003):
 1. Complexity of $\mathcal{H}$, $c(\mathcal{H})$,
 2. Training set size, N ,
 3. Generalization error, E , on new data
- As N increases, E decreases
- As $c(\mathcal{H})$ increases, first E decreases and then E increases.

Cross-Validation

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- To estimate generalization error, we need data unseen during training. We split the data as
 - Training set (50%)
 - Validation set (25%)
 - Test (publication) set (25%)
- Resampling when there is few data

Dimensions of a Supervised Learner

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1. Model:

$$g(x|\theta)$$

2. Loss function:

$$E(\theta|\mathcal{X}) = \sum_t L(r^t, g(x^t|\theta))$$

3. Optimization procedure:

$$\theta^* = \arg \min_{\theta} E(\theta|\mathcal{X})$$